SE444 Final Exam Study Guide & Cheat Sheet

Prof. Anis Koubaa — Fall 2025 — Probability, BN, Decision Theory, Planning, Game Theory

1. Probability Fundamentals

Probability Axioms

- $0 \leq P(A) \leq 1$
- $P(\text{true}) = 1$, $P(\text{false}) = 0$
- $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$

Joint & Marginal Probability

$$P(A, B) = P(A \wedge B) \quad \text{(joint)}$$

$$P(A) = \sum_b P(A, B = b) \quad \text{(marginalization)}$$

Conditional Probability

$$P(A|B) = \frac{P(A, B)}{P(B)} = \frac{P(A \wedge B)}{P(B)}$$

Chain Rule

$$P(A, B, C) = P(A) \cdot P(B|A) \cdot P(C|A, B)$$

Bayes' Rule (Most Important!)

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Where:

- $P(H)$ = **Prior** (belief before evidence)
- $P(E|H)$ = **Likelihood** (probability of evidence given hypothesis)
- $P(E)$ = **Evidence** (normalizing constant)
- $P(H|E)$ = **Posterior** (updated belief)

Total Probability

$$P(E) = \sum_h P(E|H = h) \cdot P(H = h)$$

Example: $P(\text{Disease}) = 0.01$, $P(+|\text{Disease}) = 0.95$, $P(+|\neg\text{Disease}) = 0.05$

$$P(+) = 0.95 \times 0.01 + 0.05 \times 0.99 = 0.059$$

$$P(\text{Disease}|+) = \frac{0.0095}{0.059} \approx 0.16$$

Independence

- $A \perp B$ iff $P(A, B) = P(A) \cdot P(B)$
- $A \perp B \mid C$ iff $P(A, B|C) = P(A|C) \cdot P(B|C)$

2. Bayesian Networks

Definition: A directed acyclic graph (DAG) where:

- Nodes = random variables
- Edges = direct dependencies
- Each node has a CPT (Conditional Probability Table)

Chain Rule for BNs

$$P(X_1, ..., X_n) = \prod_{i=1}^{n} P(X_i|\text{Parents}(X_i))$$

Example: For $A \to C$, $B \to C$, $C \to D$:

$$P(A, B, C, D) = P(A) \cdot P(B) \cdot P(C|A, B) \cdot P(D|C)$$

d-Separation (Key for Exams!)

Three Canonical Structures:

1. Chain: $A \to B \to C$

- $A \perp C \mid B$ (blocked when B observed)
- $A \not\perp C$ (active when B not observed)

2. Fork: $A \leftarrow B \to C$

- $A \perp C \mid B$ (blocked when B observed)
- $A \not\perp C$ (active when B not observed)

3. Collider (V-structure): $A \to B \leftarrow C$

- $A \perp C$ (blocked when B not observed)
- $A \not\perp C \mid B$ **(OPENS when B observed!)**
- Also opens if **descendant of B** observed

Explaining Away Effect

When a collider is observed, its parents become dependent. If one parent is observed, it "explains away" the effect, reducing belief in the other parent.

Example: Traffic ← Accident → Ambulance. If you see an ambulance and learn there's traffic, the traffic "explains" the ambulance, so accident becomes less likely.

Exact Inference

Enumeration: Sum over all hidden variables

$$P(Q|E) = \frac{\sum_H P(Q, H, E)}{\sum_{Q,H} P(Q, H, E)}$$

Variable Elimination:

1. Convert CPTs to factors $f(X_1, ..., X_k)$
2. Restrict factors by evidence
3. Eliminate hidden variables one by one:
   - Multiply all factors containing variable
   - Sum out the variable
4. Normalize final factor

3. Decision Theory

Expected Utility (MEU Principle)

$$EU(a) = \sum_s P(s) \cdot U(a, s)$$

Choose action $a^* = \arg\max_a EU(a)$

Example: Action "Fly" with utilities: Clear(+100), Storm(-50)

$$EU(\text{Fly}) = 0.7 \times 100 + 0.3 \times (-50) = 70 - 15 = 55$$

Value of Information (VPI)

$$VPI(E) = EU(\text{with info } E) - EU(\text{without info})$$

MDPs (Markov Decision Processes)

MDP Components: (S, A, P, R, γ)
- S = set of states
- A = set of actions
- $P(s'|s,a)$ = transition probability
- $R(s,a)$ = reward function
- γ = discount factor $(0 \leq \gamma < 1)$

Bellman Equation

$$V^*(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^*(s') \right]$$

Value Iteration Algorithm:

1. Initialize $V_0(s) = 0$ for all s
2. Repeat until convergence:

$$V_{k+1}(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} P(s'|s,a) V_k(s') \right]$$

Optimal Policy:

$$\pi^*(s) = \arg\max_a \left[R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^*(s') \right]$$

Example: Grid world with goal reward +10, step cost -1, $\gamma = 0.9$

$$V(s) = -1 + 0.9 \times (0.8 \times V(\text{goal}) + 0.2 \times V(s))$$

4. Classical Planning

STRIPS Representation
- **State:** Set of ground atoms (propositions)
- **Action:** Name(params) with:
  - Preconditions (must be true to apply)
  - Add list (effects that become true)
  - Delete list (effects that become false)
- **Closed World Assumption:** Unstated = false

Example Action:

```
Move(block, from, to):
  Pre: On(block, from), Clear(block), Clear(to)
  Add: On(block, to), Clear(from)
  Del: On(block, from), Clear(to)
```

Search Strategies:

- **Forward Search:** Start → Goal (apply actions)
- **Backward Search:** Goal → Start (regression)

5. Game Theory

Normal Form Game:
- Players: $\{1, 2, ..., n\}$
- Actions: A_i for each player
- Payoffs: $u_i(a_1, ..., a_n)$ for each player

Payoff Matrix (2-player game):

	Left	**Right**
Up	(a, b)	(c, d)
Down	(e, f)	(g, h)

First number = Row player, Second = Column player

Dominant Strategies

Strictly Dominated: Strategy s_i is dominated if $\exists s_i'$ such that $u_i(s_i') > u_i(s_i)$ for ALL opponent strategies.

IESDS: Iteratively eliminate strictly dominated strategies until no more can be removed.

Nash Equilibrium

Definition: A strategy profile $(s_1^*, s_2^*, ...)$ where NO player can improve by unilaterally changing strategy.

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i \in A_i$$

Finding NE (Best Response Method):

1. For each row, underline Column's best payoff
2. For each column, underline Row's best payoff
3. Cells with BOTH underlined = Nash Equilibrium

Example: Prisoner's Dilemma

	Cooperate	**Defect**
Cooperate	(-1, -1)	(-3, 0)
Defect	(0, -3)	**(-2, -2)**

NE = (Defect, Defect), even though (Coop, Coop) is better for both!

Key Game Types

Prisoner's Dilemma: Mutual cooperation ¿ mutual defection, but defection strictly dominates.

Coordination Game: Multiple NE, players prefer to coordinate.

Zero-Sum Game: $u_1 + u_2 = 0$ (one's gain = other's loss).

6. Project Code Concepts

Search Algorithms:

- **BFS:** FIFO queue, optimal for unit costs
- **UCS:** Priority queue by $g(n)$, optimal
- **A*:** Priority queue by $f(n) = g(n) + h(n)$
- Optimal if $h(n)$ is admissible ($h(n) \leq h^*(n)$)

Manhattan Distance:

$$h(n) = |x_n - x_g| + |y_n - y_g|$$

Admissible for 4-connected grid.

Bayesian Belief Update:

$$P(\text{obs}|\text{detect}) = \frac{P(\text{detect}|\text{obs}) \cdot P(\text{obs})}{P(\text{detect})}$$

Key Formulas Summary

Bayes' Rule: $P(H|E) = \frac{P(E|H)P(H)}{P(E)}$ **Expected Utility:** $EU(a) = \sum_s P(s)U(a,s)$ **Bellman:** $V^*(s) = \max_a[R + \gamma \sum P \cdot V^*]$